

国防科技大学数学竞赛试题

(非数学类, 2020)

考试形式: 闭卷 考试时间: 150 分钟 满分: 100 分

一、解答下列各题 (每小题 6 分, 共 30 分)

1. 设函数 $y = y(x)$ 由方程 $x = \int_0^y \frac{dt}{\sqrt{1+4t^2}}$ 确定, 求 $y''' - 4y'$.

【解】方程 $x = \int_0^y \frac{dt}{\sqrt{1+4t^2}}$ 两边同对 x 求导, 得

$$1 = \frac{y'_x}{\sqrt{1+4y^2}}, \quad \text{即 } y'_x = \sqrt{1+4y^2}, \quad (2 \text{ 分})$$

则

$$y''_{xx} = (\sqrt{1+4y^2})'_y \cdot y'_x = \frac{4y}{\sqrt{1+4y^2}} \cdot \sqrt{1+4y^2} = 4y, \quad (4 \text{ 分})$$

$$y'''_{xxx} = (4y)'_y \cdot y'_x = 4\sqrt{1+4y^2},$$

因此 $y''' - 4y' = 0$. (6 分)

2. 设悬链线方程为 $y = \frac{1}{2}(e^x + e^{-x})$, 它在 $[0, t]$ 上的曲边梯形绕 x 轴一周所得旋转

体的侧面积和 $x = t$ 处的截面面积分别记为 $S(t)$ 和 $F(t)$, 计算极限 $\lim_{t \rightarrow +\infty} \frac{S(t)}{F(t)}$.

【解】旋转体的侧面积为

$$S(t) = 2\pi \int_0^t \frac{1}{2}(e^x + e^{-x}) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \frac{\pi}{4}(e^{2t} - e^{-2t}) + t\pi, \quad (2 \text{ 分})$$

$x = t$ 处的截面面积为

$$F(t) = \pi \left[\frac{1}{2}(e^t + e^{-t}) \right]^2 = \frac{\pi}{4}(e^{2t} + e^{-2t} + 2), \quad (4 \text{ 分})$$

因此

$$\lim_{t \rightarrow +\infty} \frac{S(t)}{F(t)} = \lim_{t \rightarrow +\infty} \frac{\frac{\pi}{4}(e^{2t} - e^{-2t}) + t\pi}{\frac{\pi}{4}(e^{2t} + e^{-2t} + 2)} = \lim_{t \rightarrow +\infty} \frac{1 - e^{-4t} + 4te^{-2t}}{1 - e^{-4t} + 2e^{-2t}} = 1. \quad (6 \text{ 分})$$

3. 设 C 是椭圆柱面 $x^2 + 2y^2 = 1$ 与椭圆抛物面 $x^2 + 2y^2 = -z$ 相交而成的正向曲线, 计算曲线积分 $\int_C (x + \sqrt{2}y^3z)dx + (x - \sqrt{2}y)dy + (x + y + z)dz$.

【解】曲线 C 的参数方程为 $x = \cos \theta$, $y = \frac{1}{\sqrt{2}} \sin \theta$, $z = -1$, $\theta: 0 \rightarrow 2\pi$, (2 分)

则

$$\begin{aligned} & \int_C (x + \sqrt{2}y^3z)dx + (x - \sqrt{2}y)dy + (x + y + z)dz \\ &= \int_0^{2\pi} \left[(\cos \theta - \frac{1}{2} \sin^3 \theta)(-\sin \theta) + \frac{1}{\sqrt{2}}(\cos \theta - \sin \theta) \cos \theta \right] d\theta \\ &= \frac{1}{2} \int_0^{2\pi} \sin^4 \theta d\theta + \frac{\sqrt{2}}{2} \int_0^{2\pi} \cos^2 \theta d\theta - (\frac{\sqrt{2}}{2} + 1) \int_0^{2\pi} \sin \theta \cos \theta d\theta \quad (4 \text{ 分}) \end{aligned}$$

$$= 2 \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta + 2\sqrt{2} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} + 2\sqrt{2} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3+4\sqrt{2}}{8} \pi. \quad (6 \text{ 分})$$

4. 设 $y_1 = e^x$ 是方程 $xy'' - 2(x+1)y' + (x+2)y = 0$ 的一个特解, 求该方程的通解.

【解】令 $y = u(x)e^x$, 代入原方程, 得

$$xe^x(u'' + 2u' + u) - 2(x+1)e^x(u' + u) + (x+2)ue^x = 0,$$

因为 $e^x \neq 0$, 所以

$$x(u'' + 2u' + u) - 2(x+1)(u' + u) + (x+2)u = 0,$$

即

$$xu'' - 2u' = 0. \quad (3 \text{ 分})$$

令 $p = u'$, 则

$$x \frac{dp}{dx} - 2p = 0,$$

解得 $p = C_0 x^2$, 从而

$$\frac{du}{dx} = C_0 x^2,$$

积分得

$$u = C_1 x^3 + C_2, \text{ 其中 } C_1 = \frac{1}{3} C_0,$$

故所求方程的通解为 $y = e^x(C_1 x^3 + C_2)$. (6 分)

5. 设 n 为正整数, 求极限 $\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2}}$.

【解 1】因为 $1^2 + 2^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1)$, 所以

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{\frac{1}{6}n(n+1)(2n+1)}} \quad (2 \text{ 分})$$

$$\begin{aligned} &= \sqrt{3} \lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{n^3}} = \sqrt{3} \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{\frac{k}{n}} \cdot \frac{1}{n} \\ &= \sqrt{3} \int_0^1 \sqrt{x} dx = \frac{2\sqrt{3}}{3}. \end{aligned} \quad (6 \text{ 分})$$

【解 2】由 Stolz 定理, 得

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2} - \sqrt{1^2 + 2^2 + \cdots + (n-1)^2}} \quad (3 \text{ 分})$$

$$= \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n^2} \left(\sqrt{\frac{n(n+1)(2n+1)}{6}} + \sqrt{\frac{(n-1)n(2n-1)}{6}} \right) = \frac{2\sqrt{3}}{3}. \quad (6 \text{ 分})$$

【解 3】函数 $f(x) = \frac{1}{\sqrt{x}}$ 在 $(0, +\infty)$ 上为严格单调减函数, 则

$$\frac{1}{\sqrt{k+1}} = \int_k^{k+1} \frac{1}{\sqrt{k+1}} dx < \int_k^{k+1} \frac{1}{\sqrt{x}} dx < \int_k^{k+1} \frac{1}{\sqrt{k}} dx = \frac{1}{\sqrt{k}},$$

将 $k=1, 2, \dots, n-1$ 对应的不等式左端相加, 得

$$\sum_{k=2}^n \frac{1}{\sqrt{k}} = \sum_{k=1}^{n-1} \frac{1}{\sqrt{k+1}} < \int_1^n \frac{1}{\sqrt{x}} dx.$$

将 $k=1, 2, \dots, n$ 对应的不等式右端相加, 得

$$\int_1^{n+1} \frac{1}{\sqrt{x}} dx < \sum_{k=1}^n \frac{1}{\sqrt{k}}.$$

故

$$\frac{2}{3}[(n+1)^{\frac{3}{2}} - 1] = \int_1^{n+1} \frac{1}{\sqrt{x}} dx < \sum_{k=1}^n \frac{1}{\sqrt{k}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx = 1 + \frac{2}{3}(n^{\frac{3}{2}} - 1). \quad (3 \text{ 分})$$

因为

$$\lim_{n \rightarrow 0} \frac{\frac{2}{3}[(n+1)^{\frac{3}{2}} - 1]}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \lim_{n \rightarrow 0} \frac{\frac{2}{3}[(n+1)^{\frac{3}{2}} - 1]}{\sqrt{\frac{1}{6}n(n+1)(2n+1)}} = \frac{2\sqrt{3}}{3},$$

$$\lim_{n \rightarrow 0} \frac{1 + \frac{2}{3}(n^{\frac{3}{2}} - 1)}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \lim_{n \rightarrow 0} \frac{1 + \frac{2}{3}(n^{\frac{3}{2}} - 1)}{\sqrt{\frac{1}{6}n(n+1)(2n+1)}} = \frac{2\sqrt{3}}{3}, \quad (5 \text{ 分})$$

由夹逼准则, 得

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \frac{2\sqrt{3}}{3}. \quad (6 \text{ 分})$$

【解 4】设 $f(x) = \sqrt{x^3}$, 在区间 $[n-1, n]$ 上使用拉格朗日中值定理, 得

$$\sqrt{n^3} - \sqrt{(n-1)^3} = \frac{3}{2}\sqrt{\xi}, \quad \text{其中 } n-1 < \xi < n,$$

从而

$$\frac{3}{2}\sqrt{n-1} < \sqrt{n^3} - \sqrt{(n-1)^3} < \frac{3}{2}\sqrt{n}, \quad (2 \text{ 分})$$

取 $n \rightarrow 2, 3, \cdots, n$, 将上述 $n-1$ 个不等式相加, 得

$$\frac{3}{2}(\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n-1}) < n\sqrt{n} - 1 < \frac{3}{2}(\sqrt{2} + \sqrt{3} + \cdots + \sqrt{n}),$$

即

$$\frac{2}{3}(n\sqrt{n} - 1) + 1 < \sqrt{1} + \sqrt{2} + \cdots + \sqrt{n-1} + \sqrt{n} < \frac{2}{3}(n\sqrt{n} - 1) + \sqrt{n}, \quad (3 \text{ 分})$$

因为

$$\lim_{n \rightarrow 0} \frac{\frac{2}{3}(n\sqrt{n} - 1) + 1}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \lim_{n \rightarrow 0} \frac{\frac{2}{3}(n\sqrt{n} - 1) + 1}{\sqrt{\frac{1}{6}n(n+1)(2n+1)}} = \frac{2\sqrt{3}}{3},$$

$$\lim_{n \rightarrow 0} \frac{\frac{2}{3}(n\sqrt{n} - 1) + \sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \lim_{n \rightarrow 0} \frac{\frac{2}{3}(n\sqrt{n} - 1) + \sqrt{n}}{\sqrt{\frac{1}{6}n(n+1)(2n+1)}} = \frac{2\sqrt{3}}{3}, \quad (5 \text{ 分})$$

由夹逼准则, 得

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}}{\sqrt{1^2 + 2^2 + \cdots + n^2}} = \frac{2\sqrt{3}}{3}. \quad (6 \text{ 分})$$

二、(本题满分 10 分) 设函数 $f(x)$ 在闭区间 $[a, b]$ 上可微, 且 $f'(x) \geq f(x) > 0$,

证明不等式 $\int_a^b \frac{1}{f(x)} dx \leq \frac{1}{f(a)} - \frac{1}{f(b)}$.

【证 1】因为 $f'(x) \geq f(x) > 0$, 故 $\frac{f'(x)}{f(x)} \geq 1$, 所以

$$\frac{1}{f(x)} \leq \frac{f'(x)}{f^2(x)}, \quad (4 \text{ 分})$$

上式两边同在区间 $[a, b]$ 上积分, 得

$$\int_a^b \frac{1}{f(x)} dx \leq \int_a^b \frac{f'(x)}{f^2(x)} dx = \int_a^b \frac{df(x)}{f^2(x)} = \frac{1}{f(a)} - \frac{1}{f(b)}. \quad (10 \text{ 分})$$

【证 2】令 $F(x) = \int_a^x \frac{1}{f(t)} dt + \frac{1}{f(x)} - \frac{1}{f(a)}$, 则 $F(a) = 0$, 且 (4 分)

$$F'(x) = \frac{1}{f(x)} - \frac{f'(x)}{f^2(x)} = \frac{f(x) - f'(x)}{f^2(x)} \leq 0, \quad (7 \text{ 分})$$

故 $F(x)$ 在 $[a, b]$ 单调递减, 从而 $F(b) \leq F(a)$, 即

$$\int_a^b \frac{1}{f(x)} dx \leq \frac{1}{f(a)} - \frac{1}{f(b)}. \quad (10 \text{ 分})$$

【证 3】令 $g(x) = \ln f(x)$, 则 $f(x) = e^{g(x)}$, 由 $f'(x) \geq f(x) > 0$ 得

$$g'(x) = \frac{f'(x)}{f(x)} \geq 1, \quad (4 \text{ 分})$$

从而

$$\int_a^b \frac{1}{f(x)} dx = \int_a^b \frac{1}{e^{g(x)}} dx \leq \int_a^b \frac{g'(x)}{e^{g(x)}} dx = \frac{1}{e^{g(a)}} - \frac{1}{e^{g(b)}} = \frac{1}{f(a)} - \frac{1}{f(b)}. \quad (10 \text{ 分})$$

【证 4】由于在 $[a, b]$ 上有 $f'(x) \geq f(x) > 0$, 则

$$(e^{-x} f(x))' = e^{-x} [f'(x) - f(x)] \geq 0,$$

从而函数 $e^{-x} f(x)$ 是 $[a, b]$ 上的单调不减的正值函数, 因此当 $x \in [a, b]$ 时, 有

$$\frac{f(x)}{e^x} \geq \frac{f(a)}{e^a}, \quad \text{即} \quad \frac{1}{f(x)} \leq \frac{e^a}{e^x f(a)}, \quad (4 \text{ 分})$$

于是

$$\int_a^b \frac{1}{f(x)} dx \leq \int_a^b \frac{e^a}{e^x f(a)} dx = -\frac{e^a}{f(a)} \left(\frac{1}{e^b} - \frac{1}{e^a} \right) = \frac{1}{f(a)} \left(1 - \frac{e^a}{e^b} \right), \quad (8 \text{ 分})$$

又

$$\frac{f(b)}{e^b} \geq \frac{f(a)}{e^a}, \quad \text{即} \quad \frac{f(a)}{f(b)} \leq \frac{e^a}{e^b},$$

故

$$\int_a^b \frac{1}{f(x)} dx \leq \frac{1}{f(a)} \left(1 - \frac{e^a}{e^b} \right) \leq \frac{1}{f(a)} \left[1 - \frac{f(a)}{f(b)} \right] = \frac{1}{f(a)} - \frac{1}{f(b)}. \quad (10 \text{ 分})$$

三、(本题满分 12 分) 设函数 $f(x)$ 在 $(-\infty, +\infty)$ 上具有二阶导数, 满足 $|f(x)| \leq 1$,

$f^2(0) + [f'(0)]^2 > 1$, 证明存在 $\xi \in (-\infty, +\infty)$, 使得 $f''(\xi) + f(\xi) = 0$.

【证】令 $F(x) = f^2(x) + [f'(x)]^2$, 则 $F(0) = f^2(0) + [f'(0)]^2 = a > 1$. (2 分)

对 $f(x)$ 在 $\left[0, \frac{2}{\sqrt{a-1}}\right]$ 上使用拉格朗日中值定理, 存在 $\xi_1 \in \left(0, \frac{2}{\sqrt{a-1}}\right)$, 使得

$$|f'(\xi_1)| = \left| \frac{f\left(\frac{2}{\sqrt{a-1}}\right) - f(0)}{\frac{2}{\sqrt{a-1}}} \right| \leq \frac{\left| f\left(\frac{2}{\sqrt{a-1}}\right) \right| + |f(0)|}{\frac{2}{\sqrt{a-1}}} \leq \sqrt{a-1},$$

则

$$F(\xi_1) = f^2(\xi_1) + [f'(\xi_1)]^2 \leq 1 + (\sqrt{a-1})^2 = a,$$

同理, 存在 $\xi_2 \in \left(-\frac{2}{\sqrt{a-1}}, 0\right)$, 使得 $F(\xi_2) \leq a$. (6 分)

因为函数 $f(x)$ 在 $(-\infty, +\infty)$ 上二阶可导, 所以 $F(x)$ 在闭区间 $\left[-\frac{2}{\sqrt{a-1}}, \frac{2}{\sqrt{a-1}}\right]$ 上

连续, 从而存在 $\xi \in \left[-\frac{2}{\sqrt{a-1}}, \frac{2}{\sqrt{a-1}}\right]$, 使得 $F(\xi)$ 为 $F(x)$ 在 $\left[-\frac{2}{\sqrt{a-1}}, \frac{2}{\sqrt{a-1}}\right]$ 上

最大值, 且 $F(\xi) \geq a$. 于是, 由费马引理知,

$$F'(\xi) = 2f'(\xi)[f''(\xi) + f(\xi)] = 0. \quad (10 \text{ 分})$$

而 $[f'(\xi)]^2 = F(\xi) - f^2(\xi) \geq a - 1 > 0$, 所以 $f''(\xi) + f(\xi) = 0$. (12 分)

四、(本题满分 12 分) 设 $\Omega = \{(x, y, z) | x^2 + y^2 + z^2 \leq R^2, z \geq 0\}$ ($R > 0$), 计算三重积分 $\iiint_{\Omega} z \cos(x^2 + y^2) dx dy dz$.

【解 1】积分区域 Ω 的不等式组为 $0 \leq z \leq \sqrt{R^2 - x^2 - y^2}$, $D = \{(x, y) | x^2 + y^2 \leq R^2\}$, 由先一后二法, 得

$$\begin{aligned} \iiint_{\Omega} z \cos(x^2 + y^2) dx dy dz &= \iint_D \cos(x^2 + y^2) dx dy \int_0^{\sqrt{R^2 - x^2 - y^2}} z dz \\ &= \frac{1}{2} \iint_D (R^2 - x^2 - y^2) \cos(x^2 + y^2) dx dy \end{aligned} \quad (4 \text{ 分})$$

$$\begin{aligned} &= \frac{1}{2} \int_0^{2\pi} d\theta \int_0^R (R^2 - r^2) \cos r^2 \cdot r dr = \frac{\pi}{2} \int_0^R (R^2 - r^2) d(\sin r^2) \\ &= \frac{\pi}{2} [(R^2 - r^2)(\sin r^2)]_0^R + \pi \int_0^R r \sin r^2 dr \quad (8 \text{ 分}) \\ &= \frac{\pi}{2} \int_0^R \sin r^2 d(r^2) = \frac{\pi}{2} (1 - \cos R^2). \quad (12 \text{ 分}) \end{aligned}$$

【解 2】作球面坐标变换 $x = r \sin \varphi \cos \theta$, $y = r \sin \varphi \sin \theta$, $z = r \cos \varphi$, 则

$$\begin{aligned} \iiint_{\Omega} z \cos(x^2 + y^2) dx dy dz &= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^R r \cos \varphi \cos(r^2 \sin^2 \varphi) \cdot r^2 \sin \varphi dr \\ &= 2\pi \int_0^R dr \int_0^{\frac{\pi}{2}} r^3 \cos(r^2 \sin^2 \varphi) \sin \varphi \cos \varphi d\varphi \quad (4 \text{ 分}) \end{aligned}$$

$$\begin{aligned} &= \pi \int_0^R dr \int_0^{\frac{\pi}{2}} r^3 \cos(r^2 \sin^2 \varphi) d(\sin^2 \varphi) \\ &\quad \underline{u = \sin^2 \varphi} \quad \pi \int_0^R dr \int_0^1 r^3 \cos(r^2 u) du = \pi \int_0^R r dr \int_0^1 \cos(r^2 u) d(r^2 u) \quad (8 \text{ 分}) \\ &= \pi \int_0^R r \sin r^2 dr = \frac{\pi}{2} \int_0^R \sin r^2 d(r^2) = \frac{\pi}{2} (1 - \cos R^2). \quad (12 \text{ 分}) \end{aligned}$$

五、(本题满分 12 分)

(1) 设 $f(x)$ 在 $[0, 1]$ 上可积, 证明 $\int_0^{\frac{\pi}{2}} d\varphi \int_0^{\frac{\pi}{2}} f(\cos \varphi \cos \theta) \cos \theta d\theta = \frac{\pi}{2} \int_0^1 f(x) dx$.

(2) 计算积分 $\int_0^{\frac{\pi}{2}} d\varphi \int_0^{\frac{\pi}{2}} \frac{\sin \theta \ln(2 - \sin \theta \cos \varphi)}{2 - 2 \sin \theta \cos \varphi + \sin^2 \theta \cos^2 \varphi} d\theta$.

【证】(1) 设曲面 Σ 为球面 $x^2 + y^2 + z^2 = 1$ 第一卦限的部分, 令 $x = \cos \varphi \cos \theta$, $y = \sin \varphi \cos \theta$, $z = \sin \theta$, 由球面参数方程知 $dS = \cos \theta d\theta d\varphi$, 则

$$\int_0^{\frac{\pi}{2}} d\varphi \int_0^{\frac{\pi}{2}} f(\cos \varphi \cos \theta) \cos \theta d\theta = \iint_{\Sigma} f(x) dS. \quad (2 \text{ 分})$$

由于 Σ 的方程为 $z = \sqrt{1 - x^2 - y^2}$, 则

$$dS = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy = \frac{dx dy}{\sqrt{1 - x^2 - y^2}},$$

记 Σ 在 xOy 面上的投影区域为 $D: x^2 + y^2 \leq 1, x \geq 0, y \geq 0$, 则

$$\begin{aligned} \iint_{\Sigma} f(x) dS &= \iint_D \frac{f(x)}{\sqrt{1 - x^2 - y^2}} dx dy = \int_0^1 f(x) dx \int_0^{\sqrt{1-x^2}} \frac{1}{\sqrt{1 - x^2 - y^2}} dy \\ &= \int_0^1 f(x) \arcsin \frac{y}{\sqrt{1-x^2}} \Big|_0^{\sqrt{1-x^2}} dx = \frac{\pi}{2} \int_0^1 f(x) dx. \end{aligned} \quad (6 \text{ 分})$$

(2) 令 $\omega = \frac{\pi}{2} - \theta$, 由 (1) 知, 所求积分为

$$I = \int_0^{\frac{\pi}{2}} d\varphi \int_0^{\frac{\pi}{2}} \frac{\ln(2 - \cos \omega \cos \varphi)}{2 - 2 \cos \omega \cos \varphi + \cos^2 \omega \cos^2 \varphi} \cos \omega d\omega = \frac{\pi}{2} \int_0^1 \frac{\ln(2-x)}{2-2x+x^2} dx. \quad (8 \text{ 分})$$

令 $t = 1 - x$, 则

$$I = \frac{\pi}{2} \int_0^1 \frac{\ln(1+t)}{1+t^2} dt.$$

再令 $u = \arctan t$, 则

$$\begin{aligned} \int_0^1 \frac{\ln(1+t)}{1+t^2} dt &= \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \int_0^{\frac{\pi}{4}} \ln[1 + \tan(\frac{\pi}{4} - x)] dx \\ &= \int_0^{\frac{\pi}{4}} \ln(1 + \frac{1 - \tan x}{1 + \tan x}) dx = \int_0^{\frac{\pi}{4}} \ln \frac{2}{1 + \tan x} dx = \int_0^{\frac{\pi}{4}} \ln 2 dx - \int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx, \end{aligned}$$

所以

$$\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \frac{\pi}{8} \ln 2,$$

从而

$$I = \frac{\pi}{2} \int_0^1 \frac{\ln(1+t)}{1+t^2} dt = \frac{\pi}{2} \cdot \frac{\pi}{8} \ln 2 = \frac{\pi^2}{16} \ln 2. \quad (12 \text{ 分})$$

六、(本题满分 12 分)若数列 $\{a_n\} (n \in \mathbb{N}^+)$ 满足 $a_1 = 2$, $\sum_{k=1}^n a_k = \frac{1}{2}(n+1)(a_n+1) - 1$. 若

级数 $\sum_{n=1}^{\infty} \frac{b_n}{a_n^2}$ 收敛, 其中 $b_n \in \mathbb{R} (n=1, 2, 3, \dots)$, 计算极限 $\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \frac{b_k}{a_k}}{a_n}$.

【解】由已知 $\sum_{k=1}^n a_k = \frac{1}{2}(n+1)(a_n+1) - 1$ 得 $\sum_{k=1}^{n-1} a_k = \frac{1}{2}n(a_{n-1}+1) - 1$, 两式相减得

$$a_n = \frac{1}{2}(n+1)(a_n+1) - \frac{1}{2}n(a_{n-1}+1),$$

化简得

$$-(n-1)a_n + na_{n-1} = 1,$$

从而

$$-na_{n+1} + (n+1)a_n = 1,$$

上两式相减, 得

$$na_{n+1} - 2na_n + na_{n-1} = 0,$$

即

$$a_{n+1} + a_{n-1} = 2a_n,$$

则 $\{a_n\}$ 是等差数列. 由 $a_1 = 2$ 得 $a_2 = 3$, 故数列 $\{a_n\}$ 的通项为

$$a_n = n+1 (n \in \mathbb{N}^+). \quad (4 \text{ 分})$$

记 $S_n = \sum_{k=1}^n \frac{b_k}{(k+1)^2}$, 由于 $\sum_{n=1}^{\infty} \frac{b_n}{(n+1)^2}$ 收敛, 记 $\lim_{n \rightarrow \infty} S_n = S$, 当 $k \geq 2$ 时, 有

$$\frac{b_k}{(k+1)^2} = S_k - S_{k-1}.$$

记 $A_n = \frac{\sum_{k=1}^n \frac{b_k}{a_k}}{a_n} = \frac{1}{n+1} \sum_{k=1}^n \frac{b_k}{k+1}$, 则

$$A_n = \frac{1}{n+1} \sum_{k=1}^n (k+1) \cdot \frac{b_k}{(k+1)^2} = \frac{1}{n+1} \left[\sum_{k=2}^n (k+1)(S_k - S_{k-1}) + 2S_1 \right]$$

$$\begin{aligned}
&= \frac{1}{n+1} \left[\sum_{k=2}^n ((k+1)S_k - kS_{k-1}) - \sum_{k=2}^n S_{k-1} + 2S_1 \right] \\
&= \frac{(n+1)S_n - \sum_{k=1}^{n-1} S_k}{n+1} = S_n - \frac{n-1}{n+1} \cdot \frac{\sum_{k=1}^{n-1} S_k}{n-1}, \quad (8 \text{ 分})
\end{aligned}$$

由 $\lim_{n \rightarrow \infty} S_n = S$ 及 Stolz 公式, 得

$$\lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} S_n - \lim_{n \rightarrow \infty} \frac{n-1}{n+1} \cdot \frac{\sum_{k=1}^{n-1} S_k}{n-1} = S - \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n S_k - \sum_{k=1}^{n-1} S_k}{n - (n-1)} = 0. \quad (12 \text{ 分})$$

七、(本题满分 12 分) 设函数 $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2} (0 \leq x \leq 1)$.

(1) 证明当 $0 < x < 1$ 时有 $f(x) + f(1-x) + \ln x \ln(1-x) = \frac{\pi^2}{6}$;

(2) 计算积分 $I = \int_0^1 \frac{1}{2-x} \ln \frac{1}{x} dx$.

(1) 【证】记 $F(x) = f(x) + f(1-x) + \ln x \ln(1-x) (0 < x < 1)$, 则

$$\begin{aligned}
F'(x) &= f'(x) - f'(1-x) + \frac{\ln(1-x)}{x} + \frac{\ln x}{x-1} \\
&= \sum_{n=1}^{\infty} \frac{x^{n-1}}{n} - \sum_{n=1}^{\infty} \frac{(1-x)^{n-1}}{n} - \sum_{n=1}^{\infty} \frac{x^{n-1}}{n} + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(x-1)^{n-1}}{n} \equiv 0,
\end{aligned}$$

从而 $F(x) = C (0 < x < 1)$. 由于级数 $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}$ 的收敛半径为 $R = \lim_{n \rightarrow \infty} \sqrt[n]{n^2} = 1$,

且 $f(0) = 0$, $f(1) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, 而

$$\lim_{x \rightarrow 0^+} \ln x \ln(1-x) = \lim_{x \rightarrow 0^+} \frac{\ln(1-x)}{\frac{1}{\ln x}} = \lim_{x \rightarrow 0^+} \frac{x \ln^2 x}{1-x} = 0,$$

因此

$$C = \lim_{x \rightarrow 0^+} F(x) = \lim_{x \rightarrow 0^+} [f(x) + f(1-x) + \ln x \ln(1-x)] = \frac{\pi^2}{6},$$

即当 $0 < x < 1$ 时有 $f(x) + f(1-x) + \ln x \ln(1-x) = \frac{\pi^2}{6}$.

(2) 【解】 令 $2-x=y$ ， 则

$$I = -\int_0^1 \frac{1}{2-x} \ln x \, dx = -\int_1^2 \frac{1}{y} \ln(2-y) \, dy = -\int_1^2 \frac{\ln 2}{y} \, dy - \int_1^2 \frac{1}{y} \ln(1-\frac{y}{2}) \, dy.$$

记 $I_1 = \int_1^2 \frac{\ln 2}{y} \, dy$ ， $I_2 = \int_1^2 \frac{1}{y} \ln(1-\frac{y}{2}) \, dy$ ， 则 $I_1 = \ln^2 2$ ， 且

$$\begin{aligned} I_2 &= \int_1^2 \frac{1}{y} \ln(1-\frac{y}{2}) \, dy = -\int_1^2 \frac{1}{y} \sum_{n=1}^{\infty} \frac{y^n}{2^n \cdot n} \, dy = -\sum_{n=1}^{\infty} \int_1^2 \frac{y^{n-1}}{2^n \cdot n} \, dy \\ &= \sum_{n=1}^{\infty} \frac{1}{2^n \cdot n^2} - \sum_{n=1}^{\infty} \frac{2^n}{2^n \cdot n^2} = \sum_{n=1}^{\infty} \frac{1}{2^n \cdot n^2} - \sum_{n=1}^{\infty} \frac{1}{n^2} = f(\frac{1}{2}) - \frac{\pi^2}{6}, \end{aligned}$$

将 $x = \frac{1}{2}$ 代入 $f(x) + f(1-x) + \ln x \ln(1-x) = \frac{\pi^2}{6}$ ， 得 $f(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{\ln^2 2}{2}$ ， 所以

$$I = -\ln^2 2 + \frac{\pi^2}{6} - \frac{\pi^2}{12} + \frac{\ln^2 2}{2} = \frac{\pi^2}{12} - \frac{\ln^2 2}{2}.$$